

9. **Ans: D**

Net forces acting on satellite,

$$F_{net} = ma$$

$$G \frac{M_S m}{R^2} - G \frac{M_E m}{r^2} = ma$$

$$a = G \frac{M_S}{R^2} - G \frac{M_E}{r^2}$$

10. **Ans: C**